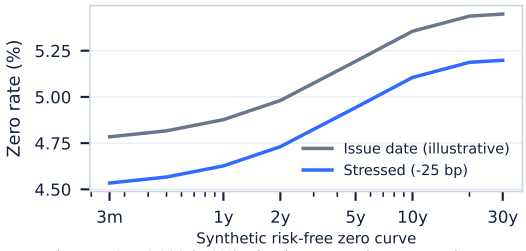
Hedging Index-Linked Insurance Liabilities Amid Equity Selloffs and Treasury Yield Volatility

As of 2026-10-16 · engine v0.1.0 · ILLUSTRATIVE: synthetic portfolio, simulated shocks, model-generated hedges — not Pacific Life positions or trading decisions

DOCUMENTED MARKET BACKDROP (DATED, SOURCED; SEE PAGE 3 FOR LINKS)

Date	Event	Source
2026-09-16	FOMC hikes 25 bp to 3.75%-4.00%, first increase in over three years	Federal Reserve press release
2026-10-07	10y yield hits 5.364% (24-year high), 30y also a 24-year high; oil above $100	Reuters
2026-10-14	Sept CPI +0.6% m/m, 3.7% y/y (core 2.6%); Oct-28 hike odds rise to ~35-54%	BLS via CNBC
2026-10-14	S&P 500 -1.1%, Nasdaq -1.9%: worst day since July; AI names lead	AP News
2026-10-14	Semis index -4.3% on the day, ~-12% in a week; NVDA -3.2%, ORCL -8.8%, DELL -11.3%	CNN Business
2026-10-14	10y yield above 5.3% intraday after CPI, closes near 5.19%; 2y ~4.81%	CNBC
2026-10-15	Nasdaq near one-month low (>4% off record); VIX 17.74 two-week high; SOXX 8th straight loss	Reuters via Kitco
2026-10-16	CoreWeave 5y CDS >1,140 bp, bonds >15%; >$1tn erased from 20 AI/semis stocks; PPI +6.3% y/y	CNBC Daily Open
2026-10-16	S&P 500 heads for worst week since June, Nasdaq >-4%; 10y quoted 5.07%-5.26%	Reuters via StreetInsider

ILLUSTRATIVE SHOCK APPLIED TO THE BLOCK (ASSUMPTION)



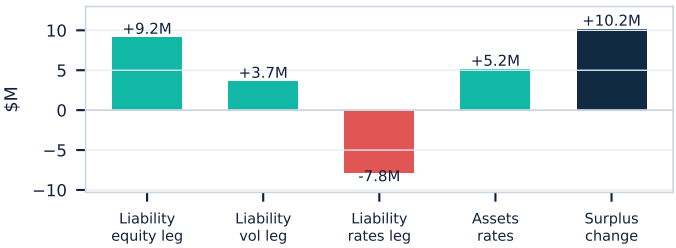
Synthetic risk-free zero curve

- Index 7,750 → 6,820 (-12%). The documented seven-session semiconductor-index drawdown applied to the broad index as an escalation; the documented broad-market move through 16 Oct was about -2% to -4%.
- 1y ATM implied vol 19% → 27% (+8 pts). The documented VIX move was about +2 points, to a two-week high of 17.74.
- Zero curve -25 bp parallel (10y 5.36% → 5.11%), inside the documented retreat from the 5.364% peak (7 Oct) to 5.07%-5.19% (15-16 Oct).

SYNTHETIC $500M BLOCK STRUCK AT 7,750 (ILLUSTRATIVE ISSUE 2026-10-06) — NOT PACIFIC LIFE POSITIONS

Product	Policies	Account value	Term	Crediting terms	PV at issue	PV stressed	Shock ΔPV
RILA	2,000	$200.0M	6y	6y buffered segment: 10% buffer, 18% cap, 100% participation, -10% floor	$158.67M	$154.37M	-$4.30M
FIA	2,500	$200.0M	10y	10y annual point-to-point: 9.5% annual cap, 85% participation, 0% floor	$175.71M	$175.30M	-$406K
IUL	1,250	$100.0M	20y	20y indexed account: 11% annual cap, 100% participation, 0% floor, 1.25% charge, 3.5% lapse, 0.3% mortality	$75.15M	$74.84M	-$310K
Total liabilities	5,750	$500.0M		Backed by a Treasury strip ladder (1y, 3y, 5y, 7y, 9y equal market value), MV = liability PV at issue, duration 5.0y vs liability 7.8y	$409.54M	$404.52M	-$5.02M

WHAT THE SHOCK DID TO THE UNHEDGED BLOCK (FULL REVALUATION, ISSUE MARKET → STRESSED MARKET, INSURER SIGN)



The unhedged block gained $10.2M. Lower index levels and higher implied vol reduced the index-linked credits owed ($12.8M of liability relief), while the -25 bp move raised the longer-duration liabilities by $7.8M against a $5.2M gain on the shorter asset ladder. The gain is one realization of a symmetric exposure: a relief rally with falling vol would have produced a loss of similar size. The hedge decision below is about removing that variance going forward, not about the direction already realized. Why the decomposition: it shows which factor the hedge must address first (equity and rates are of the same order; vol is smaller but correlated with equity).

RISK EXPOSURES AT THE STRESSED MARKET (BUMP-AND-REVALUE; ENGINE SIGN: POSITIVE = VALUE RISES)

Position	PV	Delta ($/index pt)	Vega ($/vol pt)	DV01 ($/bp)	P&L if index -10%	P&L if vol +5 pts	P&L if rates -35 bp
RILA	$154.37M	3,300	-$288K	-77,719	-$2.25M	-$1.44M	$2.72M
FIA	$175.30M	2,725	-$110K	-154,496	-$1.86M	-$551K	$5.41M
IUL	$74.84M	1,403	-$93K	-83,307	-$957K	-$464K	$2.92M
Liabilities	$404.52M	7,428	-$491K	-315,522	-$5.07M	-$2.45M	$11.04M
Assets	$414.70M	0	$0	-208,110	$0	$0	$7.28M
Net (insurer)	$10.18M	-7,428	$491K	107,412	$5.07M	$2.45M	-$3.76M

- Why these calculations: PV sets the balance sheet; delta, vega, and DV01 are the first-order sensitivities the optimizer can trade against; the last three columns translate them into first-order dollar P&L for moves of roughly one standard deviation in the stressed regime so the three risks can be compared on one scale (the Net row is what the hedge must offset).
- Reading the Net row: the insurer is short 7,428 $/pt of index delta (gains if the index falls further), long $491K per vol point (gains if implied vol rises), and long 107,412 $/bp of rates (loses $3.8M if yields fall 35 bp) because the liability duration (7.8y) exceeds the asset ladder duration (5.0y).
- The RILA's 6,975/6,200 buffer strikes are now at the money, which is where most of its delta and negative vega come from; the FIA and IUL first-year call spreads are 12% out of the money. One-month carry of the surplus is -$212K (liabilities accrete $2.0M, assets $1.8M); carry uses a one-month step because one-day theta on the Monte Carlo products is a time-grid artefact.
- Unhedged surplus P&L over the 800 out-of-sample joint shocks: standard deviation $6.1M, ES95 $14.0M. Gamma is omitted from the table: analytic for the RILA (-0.8 $/pt²) but statistically indistinguishable from zero for the Monte Carlo products at 3,000 paths (validation report §5.D).

Illustrative research exercise. The simulated shocks, the synthetic $500M portfolio, the backing assets, the hedge instruments, and every model-generated hedge are assumptions, not Pacific Life positions, hedges, or trading decisions. Documented market events are listed separately with their sources.

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HEDGE UNIVERSE PRICED FROM THE STRESSED MARKET — PREMIUM BUDGET $3.75M GROSS OPTION PREMIUM · LIQUIDITY BUDGET $10.0M (MARGIN + PREMIUM) · POSITION CAPS · COSTS

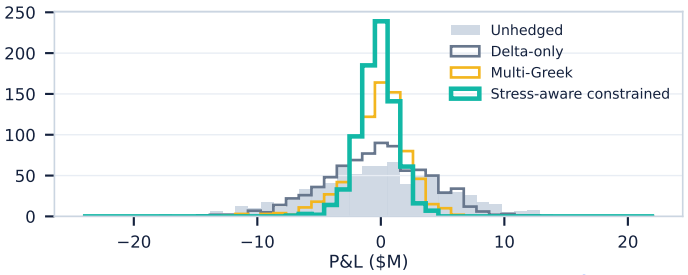
Instrument	Unit value	Premium	Liquidity / contract	Cost / contract	Delta ($/pt)	Vega ($/vol pt)	DV01 ($/bp)	Cap
Index futures (1y, 50x)	$353K	none	$35K	$15	51.7	0	35.3	±600
ATM call K=6,825 (1y, 100x)	$83K	$83K	$83K	$1,242	62.6	2,599	32.3	±300
OTM call K=7,750 (1y, 100x)	$45K	$45K	$45K	$670	43.5	2,618	23.1	±300
Put spread 6,975/6,200 (1y, 100x)	$31K	$31K	$31K	$625	-15.0	408	-13.7	±600
10y Treasury futures proxy ($100k face)	$60K	none	$2K	$25	0.0	0	-60.0	±3,500

THREE-WAY COMPARISON — SOLVED BY THE ENGINE'S CONSTRAINED OPTIMIZER, EVALUATED BY FULL REVALUATION ON 800 OUT-OF-SAMPLE JOINT SHOCKS (HIGHLIGHTED = RECOMMENDED)

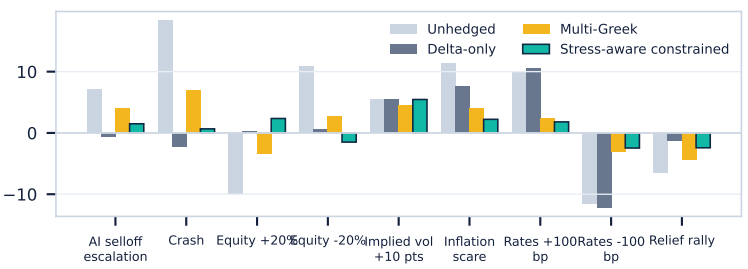
Strategy	Positions (contracts)	Premium	Txn cost	Liquidity	Greek metric	OOS var. red.	$-offset	ES95 red.	Hedged sd
Delta-only (index futures)	+144 Index futures	$0	$2K	$5.1M	16.6%	54.5%	0.47	34.4%	$4.10M
Multi-Greek (delta, vega, rho)	+134 Index futures; -38 OTM call K=7,750; -1 Put spread 6,975/6,200; +1,417 10y Treasury futures proxy	$1.73M	$64K	$10.0M	49.6%	83.9%	0.63	52.7%	$2.44M
Stress-aware constrained (scenario P&L + CVaR)	+174 Index futures; +1,543 10y Treasury futures proxy	$0	$41K	$10.0M	40.6%	93.7%	0.85	73.0%	$1.52M

Unhedged: sd $6.08M, ES95 $14.03M. 'Greek metric' is the first-order, in-sample effectiveness the dashboard reports; the validation report shows it is optimistic and not a safe optimization target. 'OOS var. red.' = 1 - Var(hedged)/Var(unhedged) and '$-offset' is the regression slope of hedge P&L on insurer loss, both on out-of-sample full revaluations — the governance-grade measures the validation recommends. Residual Greeks after hedging (insurer sign, $/pt, $/vol pt, $/bp): Delta-only +23 / +490,845 / +112,494; Multi-Greek -2,131 / +390,963 / +26,274; Stress-aware constrained +1,576 / +490,845 / +20,991 (unhedged -7,428 / +490,845 / +107,412).

Out-of-sample surplus P&L distribution (insurer sign)



Named full-revaluation stresses: surplus P&L ($M)



RECOMMENDED STRATEGY: STRESS-AWARE CONSTRAINED (SCENARIO P&L + CVAR)

- Trades (contracts): +174 Index futures (1y, 50x) (notional $61.4M); +1,543 10y Treasury futures proxy ($100k face) (notional $92.6M). No options.
- Cost: option premium $0; transaction cost $41K; liquidity (initial margin + premium) $10.00M of the $10.0M budget — the liquidity budget binds, the premium budget does not.
- Effectiveness (out-of-sample, full revaluation): variance reduction 93.7%; dollar-offset slope 0.85 (inside the 0.80-1.25 corridor; multi-Greek 0.63 and delta-only 0.47 are not); P&L sd $6.1M → $1.5M; ES95 $14.0M → $3.8M (73.0% reduction). Next best: Multi-Greek at 83.9% for $1.79M of premium and cost.
- Why it wins here: it over-hedges delta (+174 futures vs +144 for a pure delta hedge) because implied vol is assumed to rise when the index falls, so the liability's vega behaves like additional delta that futures can offset; it spends the remaining liquidity on +1,543 Treasury contracts, closing 80% of the duration gap; and it buys no options because at 27% implied vol the ATM call costs $83K per contract and each contract removes little scenario variance.
- Remaining risks: residual DV01 +20,991 $/bp (liquidity-capped: a $2.1M loss if yields fall 100 bp); residual vega $491K/pt is deliberately unhedged and becomes a loss if vol falls without the index rising; residual delta +1,576 $/pt. Worst named stress after hedging -$2.5M (Rates -100 bp) vs -$11.6M unhedged. The delta over-hedge depends on the vol-spot correlation: with uncorrelated vol the variance reduction falls to 88.0% against 86.9% for multi-Greek. Daily variation margin on +174 futures in a rally (about $3.1M per +5%) is a cash-flow risk the instantaneous scenario set does not price.
- Decision rule: highest out-of-sample variance reduction, with strategies within two points treated as equivalent and the tie broken by lower cost, then by ES95 reduction.

ROBUSTNESS — POSITIONS FIXED; OUT-OF-SAMPLE VARIANCE REDUCTION UNDER ALTERNATIVE SHOCK DISTRIBUTIONS (200 SCENARIOS EACH)

Shock distribution	Delta-only	Multi-Greek	Stress-aware constrained
Base out-of-sample set — 800 scenarios, vol beta -0.30 to the index, rates 35 bp sd independent	54.5%	83.9%	93.7%
No vol-spot correlation — Implied-vol shocks independent of the index (beta 0, noise 3.5 pts)	45.4%	86.9%	88.0%
Strong vol-spot correlation — Implied-vol beta -0.60 to the index return (noise 2.0 pts)	59.3%	76.3%	91.2%
Flight-to-quality rates — Rate shocks 60 bp sd, +0.5 correlation with the index (yields fall when equities fall)	-0.2%	84.7%	91.6%
Inflation-scare rates — Rate shocks 60 bp sd, -0.5 correlation with the index (yields rise when equities fall)	52.8%	87.8%	96.3%

Reading the comparison: the delta-only hedge is cheapest but leaves the duration gap and vega open and loses all of its benefit when rates co-move with equities. The multi-Greek hedge does not lose on sophistication grounds; it loses because the first-order Greek objective spends scarce liquidity selling calls to chase a vega match that the scenario analysis shows is worth little, and because it cannot see the vol-spot correlation. The stress-aware hedge ranks first on every distribution tested, but its margin over multi-Greek shrinks to about one point when vol is uncorrelated with the index — that assumption, not the optimizer, is what the recommendation rests on.

Illustrative research exercise. The simulated shocks, the synthetic $500M portfolio, the backing assets, the hedge instruments, and every model-generated hedge are assumptions, not Pacific Life positions, hedges, or trading decisions. Documented market events are listed separately with their sources.

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KEY ASSUMPTIONS (ILLUSTRATIVE; SEPARATED FROM THE DOCUMENTED EVENTS ON PAGE 1)

- Synthetic broad U.S. equity index at 7,750 on the illustrative issue date (6 October 2026). The level is in the neighbourhood of the documented S&P 500 prints of ~7,700-7,750 on 15-16 October but the index is synthetic.
- Pre-shock market: 1y ATM implied volatility 19% (the documented VIX, a 30-day measure, was 16-18); zero curve 4.75% + 0.70% x (1 - exp(-t/5)), anchoring the 10y near the documented 5.36% peak (2y 4.98%, 10y 5.36%); dividend yield 1.2%.
- Illustrative shock applied instantaneously to the block: equity -12% (the documented seven-session semiconductor-index drawdown applied to the broad index as an escalation scenario; the documented broad-market drawdown was roughly 2-4%), ATM implied volatility +8 points to 27% (the documented VIX move was about +2 points), parallel zero-curve shift -25 bp (inside the documented 5.36% -> 5.07%-5.19% move).
- Synthetic $500M block of newly issued RILA, FIA, and IUL segments struck at 7,750, backed by a Treasury strip ladder whose market value equalled the liability PV at issue. The index-linked credits are not yet hedged at the valuation date; the case study is the hedge-design decision at the stressed market.
- Forward-looking scenario distribution for hedge design and evaluation (roughly a one-month horizon in the stressed regime): Student-t(4) index shocks with 7.8% standard deviation, implied-volatility shocks of -0.30 x index return plus N(0, 2.5 pts) noise (clipped to [-15, +40] points), and independent N(0, 35 bp) parallel rate shocks. In-sample and out-of-sample sets use different seeds.
- Hedge budget 0.75% of account value ($3.75M) of gross option premium (premium received on sold options is counted as consumed, the engine's conservative convention); liquidity budget $10M of cash for initial margin and premium (equity futures 10% of notional, Treasury futures 2.5% of face, options at premium); position caps per instrument; transaction costs (futures $15, Treasury futures $25, options 1.5% of premium per contract).

MODELING WORKFLOW — EVERY BOX IS EXISTING, VALIDATED ENGINE CODE; THE CASE STUDY MODULE ADDS ORCHESTRATION, SCENARIOS, AND THE DECISION RULE



VALIDATION EVIDENCE RELIED ON (MODEL_VALIDATION_REPORT.MD: 126 INDEPENDENT CHECKS, 118 PASS, 0 FAIL, 8 INFORMATIONAL)

Component used here	Evidence	Status
Buffered RILA analytic value (bond + call spread − put spread)	Matches lognormal quadrature and the independent decomposition to ≤ 2e-13; replication legs reproduce the payoff to 3e-14.	PASS
FIA ratchet and IUL Monte Carlo	Within 3 standard errors of an independent closed form / deterministic recursion; forward-rate and forward-variance stepping reprices the market forward and ATM options on sloped curves.	PASS (fixed in 1b5affb)
Bump-and-revalue Greeks and DV01	Match analytic Black-Scholes Greeks; O(h²) error ratios 3.99-4.02 when halving bumps; DV01 = rho × 1e-4 to first order; cohort → portfolio aggregation exact.	PASS
Full-revaluation stresses (MarketState.bumped)	Stressed values equal independent repricing at the shocked spot/vol/rate (≤ 7e-11); shocks applied exactly as specified.	PASS
Constrained optimizer (lots, premium and liquidity budgets, CVaR)	Unconstrained solution = weighted least squares (1e-4); lot-constrained objective equals the brute-force optimum; budgets respected after rounding; reported CVaR equals independent discrete CVaR.	PASS (fixed in 700329f)
Scenario-based effectiveness instead of the Greek metric	Validation §4: the Greek metric read 1.000 where full-revaluation variance reduction was 0.94-0.95, and -9.06 for an uncapped collinear hedge; it recommends variance reduction / dollar-offset for governance.	Adopted here
Hedge sign convention	Hedge assets carry the same-signed Greeks as the liability; the optimizer receives the insurer's net exposure. Regression-tested.	PASS (fixed in 3bb67cc)
This case study	tests/test_case_study.py re-runs run_case_study() and checks that the committed JSON consumed by this PDF and by the dashboard reconciles; effectiveness metrics are recomputed with validation/reference.py; hedged stress P&L is rebuilt from trades. Runtime 60s.	Regression test

LIMITATIONS AND WHAT WOULD CHANGE THE ANSWER

- Model-consistent test: hedges are designed and evaluated with the same engine (Black-Scholes replication, GBM Monte Carlo without skew for FIA/IUL). Out-of-sample means different scenario draws, not a different model; basis, skew dynamics, and jump risk are not represented.
- Shocks are instantaneous; one-month carry is reported separately. Rebalancing between shocks, margin calls over time, and path-dependent gap risk are not simulated.
- FIA and IUL values use 3,000 antithetic paths with common random numbers, so scenario differences are smooth but absolute PVs carry Monte Carlo error of roughly 0.05%; their finite-difference gamma is statistically noise (see the validation report).
- One-day finite-difference theta on the 20-year IUL is dominated by a monthly time-grid re-discretization artefact in the engine; carry is therefore measured with a one-month step and theta is not used in the hedge design.
- The premium budget counts gross option premium (sold options consume budget), which penalizes the natural short-put-spread RILA hedge; margin on short options is approximated by premium.
- Treasury exposure is a zero-coupon proxy valued off the engine's risk-free curve; futures basis, repo, cheapest-to-deliver, and credit spreads on the backing assets are ignored. Behaviour (lapse, mortality) is applied only to IUL, and RILA/FIA are valued to maturity without decrements.
- Scenario correlations (vol beta -0.30 to the index, rates independent) are assumptions. The documented October 2026 pattern mixed yields-up/equities-down (CPI day) and yields-down/equities-down (flight to quality) sessions.
- Hedge-effectiveness figures are variance and ES95 reductions over the stated shock distribution; they are not regulatory, accounting, or realized effectiveness measures.

SOURCES FOR THE DOCUMENTED BACKDROP (RETRIEVED 16 OCTOBER 2026)

· 2026-09-16 — Federal Reserve press release: https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm
· 2026-10-07 — Reuters: https://www.reuters.com/business/us-30-year-bond-yield-hits-fresh-24-year-high-2026-10-07/
· 2026-10-14 — BLS via CNBC; Reuters: https://www.cnbc.com/2026/10/14/cpi-inflation-report-september-2026.html
· 2026-10-14 — AP News: https://apnews.com/article/stock-markets-earnings-bonds-artificial-intelligence-704d3190f0fba2908df07c85f26346ec
· 2026-10-14 — CNN Business: https://www.cnn.com/2026/10/14/investing/us-stocks-ai-chipmakers
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· 2026-10-15 — Reuters via Kitco; MarketWatch via Morningstar: https://www.kitco.com/news/off-the-wire/2026-10-15/nasdaq-slides-near-one-month-low-ai-rout-deepens
· 2026-10-16 — CNBC Daily Open: https://www.cnbc.com/2026/10/16/cnbc-daily-open-ai-inflation-coreweave-.html
· 2026-10-16 — Reuters via StreetInsider; MarketWatch; Trading Economics (FRED DGS10 is the primary series): https://www.streetinsider.com/Market+Check/Wall+Street+tumbles+as+worries+about+AI+trade+and+high+oil+prices+weigh/27195576.html